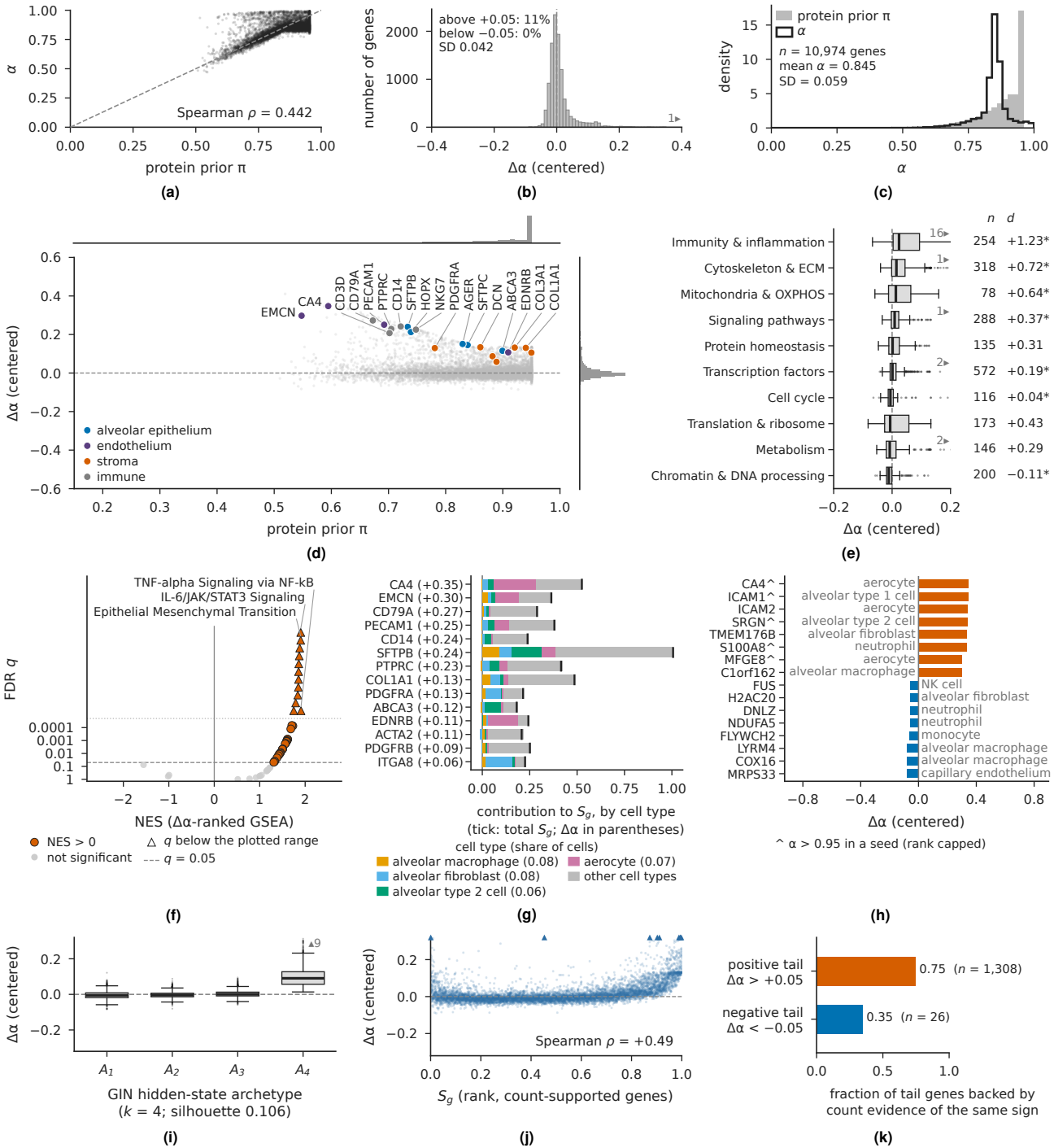

Spearman ρ = 0.442
protein prior π
α
(a)
above +0.05: 11%
below −0.05: 0%
SD 0.042
number of genes
Δα (centered)
(b)
protein prior π
α
n = 10,974 genes
mean α = 0.845
SD = 0.059
density
(c)
CA4
EMCN
CD3D
CD79A
PECAM1
PTPRC
CD14
SFTPB
HOPX
NKG7
PDGFRA
AGER
SFTPC
DCN
ABCA3
EDNRB
COL3A1
COL1A1
alveolar epithelium
endothelium
stroma
immune
Δα (centered)
protein prior π
(d)
n d
Immunity & inflammation 254 +1.23*
Cytoskeleton & ECM 318 +0.72*
Mitochondria & OXPHOS 78 +0.64*
Signaling pathways 288 +0.37*
Protein homeostasis 135 +0.31
Transcription factors 572 +0.19*
Cell cycle 116 +0.04*
Translation & ribosome 173 +0.43
Metabolism 146 +0.29
Chromatin & DNA processing 200 −0.11*
Δα (centered)
(e)
TNF-alpha Signaling via NF-kB
IL-6/JAK/STAT3 Signaling
Epithelial Mesenchymal Transition
FDR q
NES (Δα-ranked GSEA)
NES > 0
not significant
q below the plotted range
q = 0.05
(f)
CA4 (+0.35)
EMCN (+0.30)
CD79A (+0.27)
PECAM1 (+0.25)
CD14 (+0.24)
SFTPB (+0.24)
PTPRC (+0.23)
COL1A1 (+0.13)
PDGFRA (+0.13)
ABCA3 (+0.12)
EDNRB (+0.11)
ACTA2 (+0.11)
PDGFRB (+0.09)
ITGA8 (+0.06)
contribution to S_g, by cell type
(tick: total S_g; Δα in parentheses)
cell type (share of cells)
alveolar macrophage (0.08)
aerocyte (0.07)
alveolar fibroblast (0.08)
other cell types
alveolar type 2 cell (0.06)
(g)
CA4^ aerocyte
ICAM1^ alveolar type 1 cell
ICAM2 aerocyte
SRGN^ alveolar type 2 cell
TMEM176B alveolar fibroblast
S100A8^ neutrophil
MFGE8^ aerocyte
C1orf162 alveolar macrophage
FUS NK cell
H2AC20 alveolar fibroblast
DNLZ neutrophil
NDUFA5 neutrophil
FLYWCH2 monocyte
LYRM4 alveolar macrophage
COX16 alveolar macrophage
MRPS33 capillary endothelium
Δα (centered)
^ α > 0.95 in a seed (rank capped)
(h)
Δα (centered)
GIN hidden-state archetype
(k = 4; silhouette 0.106)
(i)
Spearman ρ = +0.49
S_g (rank, count-supported genes)
(j)
positive tail
Δα > +0.05
0.75 (n = 1,308)
negative tail
Δα < −0.05
0.35 (n = 26)
fraction of tail genes backed by
count evidence of the same sign
(k)